LUMS
A Not-for-Profit University

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

ASSIGNMENT 3 – SOLUTIONS

Due Date: 1 pm, November 30, 2025
Format: 10 problems for a total of 100 marks
Instructions:

- You are allowed to collaborate with your peers but copying your colleague's solution is strictly prohibited. This is not a group assignment. Each student must submit his/her own assignment.
- Solve the assignment on blank A4 sheets and staple them before submitting.
- **There may be more than one correct solution for non-numeric questions.**
- Submit in-class or in the dropbox labeled AI-501 outside the instructor's office.
- **Write your name and roll no. on the first page.**
- Feel free to contact the instructor or the teaching assistants if you have any concerns.

- You represent the most competent individuals in the country, do not let plagiarism come in between your learning. In case any instance of plagiarism is detected, the disciplinary case will be dealt with according to the university's rules and regulations.
- We require you to acknowledge any use or contributions from generative AI tools. Include the following statement to acknowledge the use of AI where applicable.

  I have used [insert Tool Name] to [write, generate, plot or compute; explain specific use of generative AI] [number of times].

Problem 1 (8 marks)

In the depths of the Anthropic Research Laboratory, Dr. Elena Chen discovers that the attention mechanism of a large language model can be represented as a Taylor polynomial. The model, codenamed Turing_n, processes text through layers where μ and σ represent learnable parameters.

Compute the Taylor polynomial T_n with $n = 0, \ldots, 5$ coefficients of the consciousness function $f(x) = \sin(x) + \cos(x)$ at the critical point $x = 0$. This represents how the model approximates its activation functions during inference.

Solution:

Step 1: General Formula The Taylor series expansion of a function $f(x)$ centered at a is:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

Given $a = 0$ (Maclaurin series), this becomes:

$$T_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

Step 2: Compute Derivatives and Evaluate at $x = 0$ We need derivatives up to $n = 5$ for $f(x) = \sin(x) + \cos(x)$.

- $n = 0$: $f(x) = \sin(x) + \cos(x) \implies f(0) = 0 + 1 = \mathbf{1}$
- $n = 1$: $f'(x) = \cos(x) - \sin(x) \implies f'(0) = 1 - 0 = \mathbf{1}$
- $n = 2$: $f''(x) = -\sin(x) - \cos(x) \implies f''(0) = 0 - 1 = \mathbf{-1}$
- $n = 3$: $f'''(x) = -\cos(x) + \sin(x) \implies f'''(0) = -1 + 0 = \mathbf{-1}$
- $n = 4$: $f^{(4)}(x) = \sin(x) + \cos(x) \implies f^{(4)}(0) = 0 + 1 = \mathbf{1}$
- $n = 5$: $f^{(5)}(x) = \cos(x) - \sin(x) \implies f^{(5)}(0) = 1 - 0 = \mathbf{1}$

Step 3: Construct the Polynomial Substituting the coefficients into the formula:

$$T_5(x) = \frac{1}{0!}x^0 + \frac{1}{1!}x^1 + \frac{-1}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5$$

Step 4: Final Simplification

$$T_5(x) = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{120}x^5$$

Problem 2 (8 marks)

Three neural architects are tasked with designing transformation functions for the embedding space of a next-generation language model. Each function maps tokens to their semantic representations.

Consider the following embedding transformations and determine the dimensional structure of each transformation's output space:

Transformation Alpha (Logarithmic Encoder):

$$f_1(x) = \sin(x_1)\cos(x_2), \quad x \in \mathbb{R}^2$$

Transformation Beta (Weighted Projection):

$$f_2(x, y) = x^\top \cdot y, \quad x, y \in \mathbb{R}^n$$

Transformation Gamma (Exponential Attention):

$$f_3(x) = xx^\top, \quad x \in \mathbb{R}^n$$

(a) What are the dimensions of $\frac{\partial f_1}{\partial x}$?

(b) Compute the Jacobians that govern gradient flow through this architecture.

Solution: (a) Dimensional Analysis:

- f_1 maps $\mathbb{R}^2 \to \mathbb{R}$ (Scalar output).
- The derivative (Jacobian) of a scalar function w.r.t a vector $x \in \mathbb{R}^2$ is a row vector of size 1×2.
- **Answer:** The dimension is 1×2.
- *Note for marking:* If a student writes 2×1, they have computed the Gradient (∇f), not the Jacobian (J). Give partial credit but note the distinction.

(b) Computation of Jacobians:

1. Transformation Alpha (f_1):

$$f_1(x) = \sin(x_1)\cos(x_2)$$

Partial derivatives:

- $\frac{\partial f_1}{\partial x_1} = \cos(x_1)\cos(x_2)$
- $\frac{\partial f_1}{\partial x_2} = \sin(x_1)(-\sin(x_2))$

$$J_{f_1} = \begin{bmatrix} \cos(x_1)\cos(x_2) & -\sin(x_1)\sin(x_2) \end{bmatrix}$$

2. Transformation Beta (f_2):

$$f_2(x, y) = x^\top y = \sum_{i=1}^{n} x_i y_i$$

This depends on two vectors x, y. The full input is $\in \mathbb{R}^{2n}$.

- Derivative w.r.t x: $\nabla_x(x^\top y) = y$. As a Jacobian row: $y^\top$.
- Derivative w.r.t y: $\nabla_y(x^\top y) = x$. As a Jacobian row: $x^\top$.

$$J_{f_2} = \begin{bmatrix} \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} y^\top & x^\top \end{bmatrix} \in \mathbb{R}^{1 \times 2n}$$

3. Transformation Gamma (f_3):

$$f_3(x) = xx^\top \quad \text{(Output is } n \times n \text{ matrix)}$$

To differentiate a matrix function w.r.t a vector, we look at element (i, j):

$$(f_3)_{ij} = x_i x_j$$

$$\frac{\partial (x_i x_j)}{\partial x_k} = \frac{\partial x_i}{\partial x_k} x_j + x_i \frac{\partial x_j}{\partial x_k} = \delta_{ik} x_j + x_i \delta_{jk}$$

Note: This is a rank-3 tensor. For the purpose of this assignment, showing the element-wise derivative above is sufficient.

Problem 3 (8 marks)

During backpropagation through a transformer's attention layers, the gradient flow encounters a nested computation path. The forward pass computes:

$$f(t) = \sin(\log(t^\top \cdot t)), \quad t \in \mathbb{R}^D$$

The attention matrix is defined as:

$$a(X) = \text{tr}(AXB)$$

Differentiate f with respect to t and a with respect to X.

Solution: Part 1: Differentiating Nested Scalar Function $f(t)$ Step 1: Decomposition Let $u = t^\top t = \|t\|^2$. (Inner scalar). Let $v = \log(u)$. (Middle scalar). Let $f = \sin(v)$. (Outer scalar).

Step 2: Chain Rule Application

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial t}$$

Step 3: Compute individual components

- $\frac{\partial f}{\partial v} = \frac{d}{dv}\sin(v) = \cos(v) = \cos(\log(t^\top t))$
- $\frac{\partial v}{\partial u} = \frac{d}{du}\log(u) = \frac{1}{u} = \frac{1}{t^\top t}$
- $\frac{\partial u}{\partial t} = \nabla_t(t^\top t) = 2t$ (This is a vector of size $D \times 1$)

Step 4: Combine

$$\nabla_t f = \big(\cos(\log(t^\top t))\big) \cdot \left(\frac{1}{t^\top t}\right) \cdot (2t)$$

$$\boxed{\nabla_t f = \frac{2t\cos(\log(t^\top t))}{t^\top t}}$$

Part 2: Differentiating Trace Function $a(X)$ Method A: Cyclic Property (Fast)

$$a(X) = \text{tr}(AXB)$$

Using the cyclic property $\text{tr}(MN) = \text{tr}(NM)$, we can permute matrices cyclically:

$$\text{tr}(AXB) = \text{tr}(BAX)$$

We use the standard identity: $\nabla_X \text{tr}(CX) = C^\top$. Let $C = BA$. Then $\nabla_X \text{tr}((BA)X) = (BA)^\top = A^\top B^\top$.

Method B: Index Notation (Rigorous)

$$a(X) = \sum_i (AXB)_{ii} = \sum_i \sum_j \sum_k A_{ij} X_{jk} B_{ki}$$

To find $\frac{\partial a}{\partial X_{pq}}$, we select only terms where $j = p$ and $k = q$:

$$\frac{\partial a}{\partial X_{pq}} = \sum_i A_{ip} B_{qi} = \sum_i B_{qi} A_{ip} = (BA)_{qp}$$

The gradient is the matrix where the (p, q) entry is $\frac{\partial a}{\partial X_{pq}}$. Thus, $\nabla_X a = ((BA)^\top)_{pq}$.

$$\boxed{\nabla_X a = A^\top B^\top}$$

Problem 4 (8 marks)

A research team at DeepMind is experimenting with novel activation functions. They need to compute all partial derivatives to understand the gradient landscape during training.

Compute the derivatives $\partial f/\partial x$ of the following activation candidates by using the chain rule. Provide the dimensions of every single partial derivative. Describe your steps in detail.

Activation Candidate A (Logarithmic Softplus):

$$f(z) = \log(1+z), \quad z = x^\top x, \quad x \in \mathbb{R}^D$$

Activation Candidate B (Generalized Sine Activation):

$$f(z) = \sin(z), \quad z = Ax + b, \quad A \in \mathbb{R}^{E\times D}, b \in \mathbb{R}^E, x \in \mathbb{R}^D$$

where $\sin(\cdot)$ is applied to every element of z.

Solution: Candidate A: Logarithmic Softplus Mapping: $x \in \mathbb{R}^D \xrightarrow{z} \mathbb{R}^1 \xrightarrow{f} \mathbb{R}^1$. **Step 1: Outer Derivative**

$$\frac{\partial f}{\partial z} = \frac{d}{dz}(\log(1+z)) = \frac{1}{1+z}$$

Dimension: Scalar (1×1).

Step 2: Inner Derivative

$$\frac{\partial z}{\partial x} = \nabla_x(x^\top x) = 2x^\top \quad \text{(Jacobian row vector)}$$

Dimension: $1 \times D$.

Step 3: Chain Rule

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = \frac{1}{1+x^\top x} \cdot 2x^\top$$

Result Dimension: $1 \times D$.

$$\boxed{\frac{\partial f}{\partial x} = \frac{2x^\top}{1+x^\top x}}$$

Candidate B: Generalized Sine Mapping: $x \in \mathbb{R}^D \xrightarrow{z} \mathbb{R}^E \xrightarrow{f} \mathbb{R}^E$. **Step 1: Outer Derivative (Jacobian of element-wise sine)** Since $f_i = \sin(z_i)$, the Jacobian is diagonal.

$$J_{f|z} = \text{diag}(\cos(z_1), \ldots, \cos(z_E)) = \text{diag}(\cos(z))$$

Dimension: $E \times E$.

Step 2: Inner Derivative (Affine transform)

$$z = Ax + b \implies \frac{\partial z}{\partial x} = A$$

Dimension: $E \times D$.

Step 3: Chain Rule

$$\frac{\partial f}{\partial x} = J_{f|z} \cdot \frac{\partial z}{\partial x}$$

$$\boxed{\frac{\partial f}{\partial x} = \text{diag}(\cos(Ax+b)) \cdot A}$$

Final Dimension Check: $(E \times E) \times (E \times D) = E \times D$. Correct.

Problem 5 (8 marks)

An AI safety researcher is analyzing the loss surface of a language model to understand training stability. The model's loss function has a complex geometric structure.

The loss function for token prediction is:

$$f(z) = \exp\left(-\frac{1}{2}z\right)$$

The pre-activation logits are computed as:

$$z = g(y) = y^\top S^{-1} y$$

The hidden representation is:

$$y = h(x) = x - \mu$$

where $\mu, x \in \mathbb{R}^D$, $S \in \mathbb{R}^{D \times D}$.

Compute the derivatives $\partial f / \partial x$ using the chain rule. Provide the dimensions of every single partial derivative. Describe your steps in detail.

Solution: Goal: Compute $\nabla_x f$. By Chain Rule: $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial z}\frac{\partial z}{\partial y}\frac{\partial y}{\partial x}$.

Step 1: Derivative of f w.r.t z

$$f(z) = e^{-\frac{1}{2}z} \implies \frac{\partial f}{\partial z} = -\frac{1}{2}e^{-\frac{1}{2}z}$$

Dimension: 1×1 (Scalar).

Step 2: Derivative of z w.r.t y

$$z = y^\top S^{-1} y$$

Using the identity $\nabla_y (y^\top A y) = (A + A^\top)y$. Here $A = S^{-1}$.

$$\nabla_y z = (S^{-1} + (S^{-1})^\top)y$$

Note: If S is a covariance matrix, it is symmetric, so S^{-1} is symmetric, simplifying to $2S^{-1}y$. However, in general form:

$$\frac{\partial z}{\partial y} = y^\top (S^{-1} + S^{-\top}) \quad \text{(Jacobian row vector)}$$

Dimension: $1 \times D$.

Step 3: Derivative of y w.r.t x

$$y = x - \mu \implies \frac{\partial y}{\partial x} = I_D \quad \text{(Identity matrix)}$$

Dimension: $D \times D$.

Step 4: Final Multiplication

$$\frac{\partial f}{\partial x} = \left[-\frac{1}{2}e^{-\frac{1}{2}z}\right] \cdot \left[y^\top (S^{-1} + S^{-\top})\right] \cdot I_D$$

Substitute $y = x - \mu$:

$$\boxed{\frac{\partial f}{\partial x} = -\frac{1}{2}\exp\left(-\frac{1}{2}(x-\mu)^\top S^{-1}(x-\mu)\right)(x-\mu)^\top (S^{-1} + S^{-\top})}$$

Final Dimension: $1 \times D$ (Gradient row vector).

Problem 6 (8 marks)

You are an AI researcher analyzing the loss surface of a neural network. The model's loss function over a single parameter space is given by:

$$f(x) = x^3 + 6x^2 - 3x - 5$$

This represents a simplified view of how loss changes as we adjust one critical hyperparameter during training.

Find the stationary points of this loss landscape and classify each one. Determine whether they represent:

- Local minima (stable training points where the model converges)
- Local maxima (unstable peaks to avoid)
- Saddle points (treacherous plateaus that slow down training)

Solution: Step 1: Compute First Derivative (Gradient)

$$f'(x) = \frac{d}{dx}(x^3 + 6x^2 - 3x - 5) = 3x^2 + 12x - 3$$

Step 2: Solve for Stationary Points ($f'(x) = 0$)

$$3x^2 + 12x - 3 = 0$$

Divide by 3:

$$x^2 + 4x - 1 = 0$$

Using Quadratic Formula $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$:

$$x = \frac{-4 \pm \sqrt{16 - 4(1)(-1)}}{2} = \frac{-4 \pm \sqrt{20}}{2} = \frac{-4 \pm 2\sqrt{5}}{2}$$

Stationary points: $x_1 = -2 + \sqrt{5}$ and $x_2 = -2 - \sqrt{5}$.

Step 3: Compute Second Derivative (Hessian)

$$f''(x) = \frac{d}{dx}(3x^2 + 12x - 3) = 6x + 12$$

Step 4: Classification (Second Derivative Test)

- **Test point $x_1 = -2 + \sqrt{5}$:**

  $$f''(x_1) = 6(-2 + \sqrt{5}) + 12 = -12 + 6\sqrt{5} + 12 = 6\sqrt{5}$$

  Since $6\sqrt{5} \approx 13.4 > 0$, the function is concave up. **Conclusion:** x_1 is a **Local Minimum**.

- **Test point $x_2 = -2 - \sqrt{5}$:**

  $$f''(x_2) = 6(-2 - \sqrt{5}) + 12 = -12 - 6\sqrt{5} + 12 = -6\sqrt{5}$$

  Since $-6\sqrt{5} \approx -13.4 < 0$, the function is concave down. **Conclusion:** x_2 is a **Local Maximum**.

Problem 7 (8 marks)

A team at DeepMind is investigating the geometric properties of feasible regions in their optimization problems. They've encountered three controversial claims about convex sets.

Determine whether the following statements are TRUE or FALSE. Provide rigorous proofs or counterexamples.

(a) The Intersection Theorem: "The intersection of any two convex sets is convex."

(b) The Union Conjecture: "The union of any two convex sets is convex."

(c) The Difference Dilemma: "The difference of a convex set A from another convex set B is convex."

Solution: Definition of Convex Set: A set S is convex if for any $x, y \in S$ and any $\theta \in [0, 1]$, the point $\theta x + (1 - \theta)y \in S$.

(a) TRUE. *Proof:* 1. Let A and B be convex sets. Let $C = A \cap B$. 2. Pick any two points $x, y \in C$. 3. By definition of intersection, $x, y \in A$ AND $x, y \in B$. 4. Since A is convex, the line segment connecting them $z = \theta x + (1 - \theta)y$ must belong to A. 5. Since B is convex, the same line segment z must belong to B. 6. Since $z \in A$ and $z \in B$, then $z \in A \cap B$. 7. Therefore, C is convex.

(b) FALSE. *Reasoning:* The union creates a shape that may not contain the line connecting two points from the separate sets. *Counterexample:* Let $A = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ (Interval $[0, 1]$). Let $B = \{x \in \mathbb{R} \mid 2 \leq x \leq 3\}$ (Interval $[2, 3]$). Union $C = [0, 1] \cup [2, 3]$. Pick $x = 1 \in C$ and $y = 2 \in C$. Let $\theta = 0.5$. The convex combination is 1.5. $1.5 \notin [0, 1]$ and $1.5 \notin [2, 3]$, so $1.5 \notin C$. Thus, C is not convex.

(c) FALSE. *Reasoning:* Subtracting a convex set (like creating a hole) usually destroys convexity. *Counterexample:* Let A be a solid unit disk in $\mathbb{R}^2$ (Convex). Let B be a smaller disk of radius 0.5 centered at the origin (Convex). The difference $D = A \setminus B$ is an annulus (a donut shape). Pick a point on the "left" side of the ring $x = (-0.8, 0)$ and a point on the "right" side $y = (0.8, 0)$. The midpoint is $(0, 0)$. However, $(0, 0) \in B$, so $(0, 0) \notin A \setminus B$. The line segment leaves the set D, so D is not convex.

Problem 8 (12 marks)

The ML operations team at Cohere is training a large classification model using the hinge loss (the loss function used by support vector machines):

$$L(\alpha) = \max(0, 1 - \alpha)$$

However, they want to apply gradient-based methods like L-BFGS that require smooth, differentiable objectives. The problem? The hinge loss has a kink at $\alpha = 1$.

(a) Compute the convex conjugate of the hinge loss $L^*(\beta)$, where β is the dual variable. Add a proximal term and compute the conjugate of:

$$L^*(\beta) + \frac{\gamma}{2}\beta^2$$

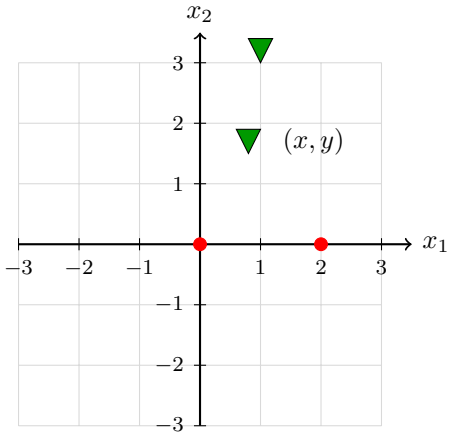
where γ is a given hyperparameter that controls smoothing strength.

(b) Explain how this transformation enables the use of subgradient-free methods while approximating the original hinge loss.

Solution:

Problem 9 (16 marks)

Consider the following 2-class classification problem with circle class as the positive class and triangle class as the negative class. There are a total of 4 points; however, one of the points is labeled as (x, y).



(a) Determine the range of values for x and y for which the data remains linearly separable.

(b) Now suppose $x = y = 1$:

(i) Draw the maximum margin separator on the graph.

(ii) Clearly label Support Vectors.

(iii) Find bias w_0 and orientation vector $w = [w_1, w_2]$ for the maximum margin separator.

(c) Now suppose $x = 1$ and $y = 4$:

(i) Draw the maximum margin separator on the graph.

(ii) Clearly label Support Vectors.

(iii) Find bias w_0 and orientation vector $w = [w_1, w_2]$ for the maximum margin separator.

Solution:

(a) **Analysis:** Two classes are linearly separable if and only if their convex hulls (shape formed by connecting the points) do not intersect.

- **Positive Class (C_+):** Points at $(0, 0)$ and $(2, 0)$. The convex hull is the line segment on the x_1-axis:

$$H_+ = \{(x_1, 0) \mid 0 \leq x_1 \leq 2\}$$

- **Negative Class (C_-):** Points at $T_1(1, 3)$ and $T_2(x, y)$. The convex hull is the line segment connecting these two points.

For the classes to be separable, the segment connecting $T_1(1, 3)$ and $T_2(x, y)$ must not intersect the segment H_+ (the interval $[0, 2]$ on the x-axis).

Case 1: $y > 0$ If the variable point has a positive y-coordinate, the entire negative class segment lies strictly above the x-axis ($x_2 > 0$). Since the positive class lies on $x_2 = 0$, there is no intersection.

$$\text{Condition 1: } y > 0 \quad \text{(valid for all } x\text{)}$$

Case 2: $y \leq 0$ If the variable point lies on or below the x_1-axis, the segment connecting $T_1(1, 3)$ to $T_2(x, y)$ crosses the line $x_2 = 0$. To maintain separability, this crossing point must occur **outside** the interval $[0, 2]$.

We define the boundaries of the "forbidden cone" by drawing lines from $T_1(1, 3)$ through the endpoints of the positive class:

(a) **Left Boundary Line:** Passes through $(1, 3)$ and $(0, 0)$.

$$m = \frac{3 - 0}{1 - 0} = 3 \implies y = 3x_1 \implies x_1 = \frac{y}{3}$$

(b) **Right Boundary Line:** Passes through $(1, 3)$ and $(2, 0)$.

$$m = \frac{3-0}{1-2} = -3 \implies y - 0 = -3(x_1 - 2) \implies y = -3x_1 + 6$$

The data is **not** separable if $T_2(x, y)$ falls in the region where the segment to T_1 crosses the interval $[0, 2]$. This occurs if $y \leq 0$ and the point lies between the two boundary lines (i.e., $y \leq 3x$ and $y \leq 6 - 3x$).

Therefore, for the data to be **separable** when $y \leq 0$, the point must lie outside this region:

$$y > 3x \quad \text{OR} \quad y > 6 - 3x$$

Final Answer: The data is linearly separable if:

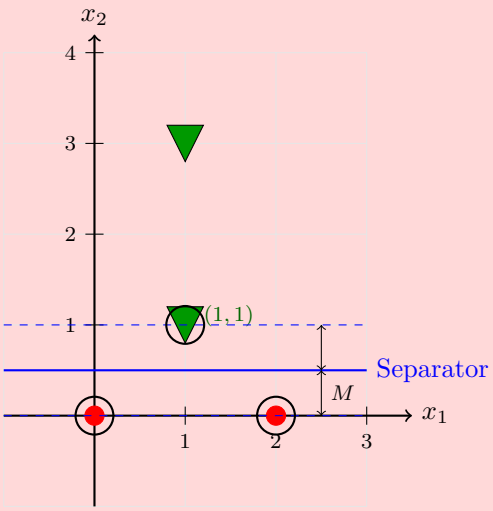
$$y > 0 \quad \cup \quad \{y \leq 0 \mid y > 3x \text{ or } y > 6 - 3x\}$$

(b) **Given:** The variable point is located at $(x, y) = (1, 1)$.

(i) & (ii) **Maximum Margin Separator and Support Vectors**

By inspecting the geometry:

- The positive class points are at $y = 0$.
- The closest negative class point is at $(1, 1)$ (since the other negative point is at $(1, 3)$).
- The maximum margin separator lies exactly in the middle of the vertical gap between $y = 0$ and $y = 1$.
- The separator is the horizontal line $x_2 = 0.5$.



(iii) **Calculation of Bias and Orientation Vector**

We assume the canonical form of the hyperplane equation: $w^T x + w_0 = \text{score}$. For Support Vectors (SVs), the score must be exactly $+1$ for the positive class and -1 for the negative class.

From the diagram, the separator is horizontal, implying $w_1 = 0$. The vector takes the form $w = [0, w_2]$.

1. Using a Positive Support Vector $(0, 0)$**:**

$$w \cdot [0, 0]^T + w_0 = +1 \implies \boldsymbol{w_0 = 1}$$

2. Using the Negative Support Vector $(1, 1)$**:**

$$w \cdot [1, 1]^T + w_0 = -1$$
$$(0)(1) + w_2(1) + 1 = -1$$
$$w_2 + 1 = -1 \implies \boldsymbol{w_2 = -2}$$

Result: The parameters for the maximum margin separator are:

$$w_0 = 1, \quad w = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

Check: The equation is $1 - 2x_2 = 0 \implies x_2 = 0.5$. This matches the graph.

(c) **Given:** The variable point is located at $(x, y) = (1, 4)$.

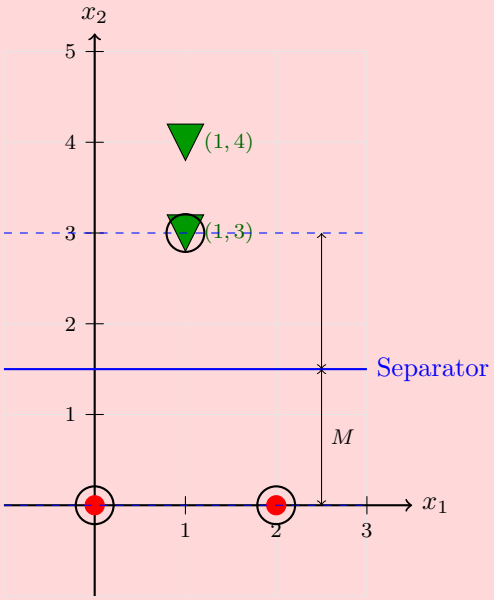
(i) & (ii) **Maximum Margin Separator and Support Vectors**

By inspecting the geometry:

- The positive class points are located on the line $x_2 = 0$.
- The negative class consists of $T_1(1,3)$ and the new point $T_2(1,4)$.
- The negative point closest to the positive class is $T_1(1,3)$. The point $T_2(1,4)$ is farther away and does not constrain the margin.
- The maximum margin separator lies halfway between the closest points of the two classes ($y = 0$ and $y = 3$).
- The separator is the horizontal line $x_2 = 1.5$.

Support Vectors:

- Positive: $(0,0)$ and $(2,0)$.
- Negative: $(1,3)$.
- Note: $(1,4)$ is **not** a support vector.



(iii) **Calculation of Bias and Orientation Vector**

We find the canonical hyperplane $w^T x + w_0 =$ score where SVs yield ± 1.
Since the boundary is horizontal, $w_1 = 0$. The vector is $w = [0, w_2]$.

1. Using a Positive Support Vector $(0,0)$**:**

$$w \cdot [0,0]^T + w_0 = +1 \implies \boldsymbol{w_0 = 1}$$

2. Using the Negative Support Vector $(1,3)$**:**

$$w \cdot [1,3]^T + w_0 = -1$$
$$(0)(1) + w_2(3) + 1 = -1$$
$$3w_2 = -2 \implies \boldsymbol{w_2 = -\frac{2}{3}}$$

Result: The parameters for the maximum margin separator are:

$$w_0 = 1, \quad w = \begin{bmatrix} 0 \\ -\frac{2}{3} \end{bmatrix}$$

Check: The equation is $1 - \frac{2}{3}x_2 = 0 \implies \frac{2}{3}x_2 = 1 \implies x_2 = 1.5$. This matches the graph.

Problem 10 (16 marks)

In the following figure, a 2-class (circle (-1) or triangle ($+1$)) classification is being considered based on two input features x_1 and x_2. Clearly, the data is not linearly separable by a single linear classifier. Our goal is to obtain a new representation of the original 2-dimensional data such that the new data becomes linearly separable by a single linear classifier in the new domain. For this purpose, perform the following:

(a) Using the given data points, fit a linear classifier of the form:

$$w^T x + w_0 = 0$$

using the average Hinge-loss over all points:

$$I(w) = \frac{1}{N} \sum_{i=1}^{N} \max(0, 1 - m_i)$$

where $m_i = y_i(w^T x_i + w_0)$ is the margin.

You may want to write a small piece of code for Gradient Descent to obtain the final weight vector. Report the training accuracy of this classifier (specifically, how many points are misclassified?).

(b) Now, use the following feature map (or kernel) to transform every 2-dimensional point x_i into a new 3-dimensional vector $\hat{\mathbf{x}}_i$:

$$\hat{\mathbf{x}}_i = \begin{bmatrix} x_{i1}^2 \\ x_{i2}^2 \\ \sqrt{2} x_{i1} x_{i2} \end{bmatrix}$$

Once you have the new data, fit a linear classifier (with one extra dimension) on the transformed data:

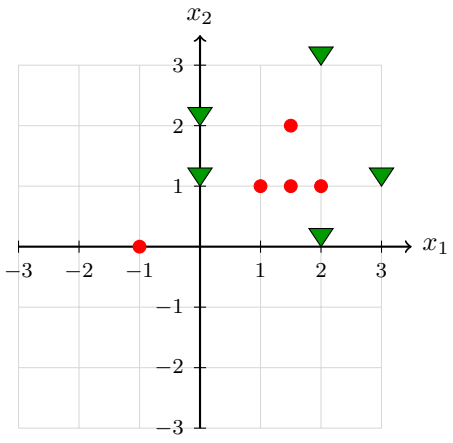
$$w^T \hat{\mathbf{x}} + w_0 = 0$$

using the average Hinge-loss over all points:

$$\tilde{I}(w) = \frac{1}{N} \sum_{i=1}^{N} \max(0, 1 - \hat{m}_i)$$

where $\hat{m}_i = y_i(w^T \hat{\mathbf{x}}_i + w_0)$.

You may want to write a small piece of code for Gradient Descent to obtain the final weight vector. Report the training accuracy of this classifier (specifically, how many points are misclassified?).



Solution:

(a) The average Hinge-loss function is defined as:

$$L(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}\max(0, 1 - m_i)$$

The gradient $\nabla_{\mathbf{w}} L(\mathbf{w})$ is calculated as:

$$\nabla_{\mathbf{w}} L(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}\begin{cases} 0 & \text{for } m_i \geq 1 \\ -y_i \mathbf{x}_i & \text{for } m_i < 1 \end{cases}$$

Final Weights Vector

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} -0.558 \\ +0.443 \\ +0.227 \end{bmatrix}$$

Predictions Table

$\mathbf{x}$ (Original)	$\mathbf{y}$ (Class)	$\hat{\mathbf{y}}$ (Pred)
$\begin{bmatrix} 0 & 1 \end{bmatrix}^T$	Δ	$\bigcirc\times$
$\begin{bmatrix} 0 & 2 \end{bmatrix}^T$	Δ	$\bigcirc\times$
$\begin{bmatrix} 2 & 0 \end{bmatrix}^T$	Δ	$\Delta\checkmark$
$\begin{bmatrix} 3 & 1 \end{bmatrix}^T$	Δ	$\Delta\checkmark$
$\begin{bmatrix} 2 & 3 \end{bmatrix}^T$	Δ	$\Delta\checkmark$
$\begin{bmatrix} -1 & 0 \end{bmatrix}^T$	$\bigcirc$	$\bigcirc\checkmark$
$\begin{bmatrix} 1.5 & 1 \end{bmatrix}^T$	$\bigcirc$	$\Delta\times$
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	$\bigcirc$	$\Delta\times$
$\begin{bmatrix} 1.5 & 2 \end{bmatrix}^T$	$\bigcirc$	$\Delta\times$
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	$\bigcirc$	$\Delta\times$

Result

$$\boxed{\text{Accuracy} = \frac{4}{10} = 0.4}$$

(b) Using Kernel:

$$\hat{x} = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}$$

Data Transformation Table

$\mathbf{x}$ (Original)	$\hat{\mathbf{x}}$ (Transformed)	$\mathbf{y}$ (Class)	$\hat{\mathbf{y}}$ (Pred)
$\begin{bmatrix}0 & 1\end{bmatrix}^T$	$\begin{bmatrix}0 & 1 & 0\end{bmatrix}$	Δ	$\bigcirc\times$
$\begin{bmatrix}0 & 2\end{bmatrix}^T$	$\begin{bmatrix}0 & 4 & 0\end{bmatrix}$	Δ	$\Delta\checkmark$
$\begin{bmatrix}2 & 0\end{bmatrix}^T$	$\begin{bmatrix}4 & 0 & 0\end{bmatrix}$	Δ	$\Delta\checkmark$
$\begin{bmatrix}3 & 1\end{bmatrix}^T$	$\begin{bmatrix}9 & 1 & 3\sqrt{2}\end{bmatrix}$	Δ	$\Delta\checkmark$
$\begin{bmatrix}2 & 3\end{bmatrix}^T$	$\begin{bmatrix}4 & 9 & 6\sqrt{2}\end{bmatrix}$	Δ	$\Delta\checkmark$
$\begin{bmatrix}-1 & 0\end{bmatrix}^T$	$\begin{bmatrix}1 & 0 & 0\end{bmatrix}$	$\bigcirc$	$\bigcirc\checkmark$
$\begin{bmatrix}1 & 1\end{bmatrix}^T$	$\begin{bmatrix}1 & 1 & \sqrt{2}\end{bmatrix}$	$\bigcirc$	$\bigcirc\checkmark$
$\begin{bmatrix}1.5 & 1\end{bmatrix}^T$	$\begin{bmatrix}2.25 & 1 & \frac{3\sqrt{2}}{2}\end{bmatrix}$	$\bigcirc$	$\bigcirc\checkmark$
$\begin{bmatrix}2 & 1\end{bmatrix}^T$	$\begin{bmatrix}4 & 1 & 2\sqrt{2}\end{bmatrix}$	$\bigcirc$	$\bigcirc\checkmark$
$\begin{bmatrix}1.5 & 2\end{bmatrix}^T$	$\begin{bmatrix}2.25 & 4 & 3\sqrt{2}\end{bmatrix}$	$\bigcirc$	$\bigcirc\checkmark$

Final Weights Vector

$$\mathbf{W} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -1.569 \\ 0.7025 \\ 1.053 \\ -1.14834141 \end{bmatrix}$$

Result

$$\boxed{\text{Accuracy} = \frac{9}{10} = 0.9}$$

— End of Assignment —

Question	Points	Score
1	8	
2	8	
3	8	
4	8	
5	8	
6	8	
7	8	
8	12	
9	16	
10	16	
Total:	100	